

Rohan Khataavkar

PhD Student in Mechanical Engineering

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Education

PhD	Arizona State University , Mechanical Engineering	Tempe, Arizona, USA
	- Expected graduation: December 2028 - CGPA: 4/4	Jan 2024 – Dec 2028
Master of Technology	College of Engineering Pune, Savitribai Phule Pune University , Mechanical Engineering	Pune, India
	CGPA: 9.51/10	Aug 2021

Academic Research Experience

Arizona State University, School for Engineering of Matter, Transport and Energy , Graduate Research Associate	Tempe, Arizona, USA
Advisors: Dr. Jiefeng Sun and Dr. Hyunglae Lee - Soft Tunable Passive Back Support Device: Developed a task-oriented soft passive device with tunable stiffness (0.58, 0.92, and 1.7 N/mm) and slack length (0-67 mm), combining assistance tuning with a compact passive design. - Demonstrated the device's efficacy in helping users regain balance after slip perturbations during standing and walking through a biomechanics study. - Soft Semi-Active Back Support Device: Combined a variable-stiffness passive element with a pneumatic artificial muscle in parallel, enabling external energy addition with a compact and lightweight device. - Demonstrated reductions in human back-muscle effort through an electromyography study.	Jan 2024 – present
Indian Institute of Technology Delhi, Centre for Biomedical Engineering , Research Fellow	New Delhi, India
Advisor: Dr. Deepak Joshi - Haptic Cueing System for Parkinson's Disease: Formulated and pilot-tested a kinematic gait parameter-driven vibrotactile cueing system for freezing-of-gait mitigation and toe-clearance enhancement. - The system reduced freezing severity in 8 of 10 participants with Parkinson's disease and increased toe clearance in 7 of 8 participants. - Force Myography-based User Intent Detection: Developed FMG bands and decision-tree algorithms that detected step-length change intent with an F1 score of 86.64%. - Cable-driven Knee Exoskeleton: Built a system for classifying people with osteoarthritis by Kellgren-Lawrence grade. - Powered Ankle-Foot Prosthesis: Developed a prosthesis for assisting amputees with weight-shifting exercises during early post-amputation rehabilitation.	Oct 2021 – Oct 2023
College of Engineering Pune , Master's Thesis Researcher	Pune, India
Advisor: Dr. Ajay Bhattu - Synthesized and optimized the fatigue life of a spring-less compliant quadruped robot leg using inherent elastodynamics and Abaqus CAE. - Reduced the leg assembly from five parts to two while maintaining kinematic and dynamic equivalence with a spring-loaded rigid linkage.	Aug 2020 – Aug 2021

Industrial Research and Development Experience

Garrett Motion, R&D Intern

Aug 2021 – Sept 2021

- Designed a torque-to-yield bolted joint to maintain joint integrity in high-pressure compressors at elevated temperatures.

American Axle and Manufacturing, R&D Intern

Aug 2017 – May 2018

- Incorporated an electronic locking feature into an existing open differential while minimizing changes to the original assembly and built a functional prototype.
- Packaged the electromagnetic actuator and locking mechanism on opposite sides of the ring gear to fit within the existing compact carrier and axle housing.
- Used HyperMesh, Abaqus, and selective laser sintering for analysis and prototyping.
- Developed an automatic differential locking and unlocking algorithm in MATLAB and Simulink.

College Racing Team, R&D Staff

Aug 2015 – Oct 2016

- SAE EffiCycle 2016: Designed a chain-and-cassette transmission, reduction gearbox, and differential gearbox; calculated drivetrain performance; and supported CAD, assembly, and GD&T drafting.
- SAE Supra 2016: Designed the engine intake manifold and simulated airflow through a venturi and orifice using Abaqus CAE.

Awards and Fellowships

Jumpstart Research Grant

2025

\$1,500 research grant from Arizona State University, awarded in Spring and Fall 2025.

Senior Research Fellowship

Oct 2021 – Sept 2023

\$10,895.92 fellowship from the Indian Institute of Technology Delhi.

AICTE Post Graduate Scholarship

Aug 2019 – July 2021

\$3,386.20 scholarship from the All India Council for Technical Education.

Bravo Award

Sept 2021

Garrett Motion.

Graduate Aptitude Test in Engineering (GATE)

Mar 2019

Secured the 98.22 percentile in the national master's entrance examination.

Publications

A Hybrid Variable-Stiffness Soft Back Support Device

2025

Rohan Khataavkar, The Bach Nguyen, Yuanhao Chen, Hyunglae Lee, Jiefeng Sun
[10.1109/LRA.2025.3541459](#) (IEEE Robotics and Automation Letters)

A Well-Tuned HARP: Versatile Artificial Muscle by Decoupling Anisotropy

2025

Eric Weissman, Rohan Khataavkar, Jiefeng Sun
(Proceedings of the National Academy of Sciences of the United States of America (under review in the uploaded CV))

A Semi-Active Soft Back Support Device

2026

Rohan Khataavkar, The Bach Nguyen, Inseung Kang, Hyunglae Lee, Jiefeng Sun
(In preparation)

A Novel Kinematic Gait Parameter-Based Vibrotactile Cue for Freezing of Gait Mitigation Among Parkinson's Patients: A Pilot Study Rohan Khataavkar, Ashutosh Tiwari, Priyanka Bhat, Achal Srivastava, S. Senthil Kumaran, Deepak Joshi 10.1109/TOH.2024.3378917 (IEEE Transactions on Haptics)	2024
Effect of a Kinematic Gait Parameter-Based Haptic Cue on Toe Clearance in Parkinson's Patients Rohan Khataavkar, Ashutosh Tiwari, Priyanka Bhat, Deepak Joshi 10.1007/s10439-024-03501-4 (Annals of Biomedical Engineering)	2024
Synthesis and Fatigue Life Optimization of a Spring-Less Compliant Robot Leg Design Rohan Vijay Khataavkar, Ajay Pandit Bhattu 10.1007/978-981-19-3938-9_48 (National Conference on Multidisciplinary Analysis and Optimization)	2023
Gait Step Length Classification Using Force Myography Rohan Khataavkar, Ashutosh Tiwari, Rishabh Bajpai, Deepak Joshi 10.1109/ICONAT53423.2022.9726014 (2022 International Conference for Advancement in Technology (ICONAT))	2022
Exploring the Center of Pressure Shift Feedback at Heel Strike to Modulate the Step Length Ashutosh Tiwari, Rishabh Bajpai, Rohan Khataavkar, Manish Gupta, Bhavuk Garg, Deepak Joshi 10.1109/ICONAT53423.2022.9726031 (2022 International Conference for Advancement in Technology (ICONAT))	2022
AbnormNet: A Neural Network Based Suggestive Tool for Identifying Gait Abnormalities in Cerebral Palsy Children Rishabh Bajpai, Ashutosh Tiwari, Deepak Joshi, Rohan Khataavkar 10.1109/ICONAT53423.2022.9725832 (2022 International Conference for Advancement in Technology (ICONAT))	2022

Skills

Tools: MATLAB, PyTorch, OpenSim, Arduino, Raspberry Pi, Vicon Nexus, SolidWorks, Fusion 360, Siemens NX, Abaqus CAE, Ansys, SMC PneuDraw, Overleaf, Adobe Illustrator, Adobe Premiere Pro

Design: Mechatronics and robotic system design, pneumatic system design, CAD, additive manufacturing, solid mechanics, finite-element analysis

Experimental: Instron force testing, Vicon optical motion capture, Delsys EMG acquisition and processing, Speedgoat real-time multi-sensor capture, movement biomechanics, signal processing, clinical protocol formulation, patient testing

Extracurricular

College of Engineering Pune, Graduate Assistant

Assisted the Admissions Committee in evaluating applicants to the master's program in mechanical engineering.

Professional Service

Reviewer: Scientific Reports, IEEE Sensors Journal, IEEE Robotics and Automation Letters, IEEE RoboSoft, IEEE/ASME AIM, American Control Conference

Professional Memberships

Memberships: Tau Beta Pi; IEEE (student member)